

- 1 (a) The equation of a progressive wave is given as $y = 0.2 \sin(100\pi t + 4\pi x)$ where x and y in cm, and t in s. Determine in **SI unit** the
- (i) frequency.
 - (ii) wave number.
 - (iii) wavelength.
 - (iv) vibrational velocity of a particle at position $x = 0.15$ cm at time $t = 0.2$ s.
 - (v) wave propagation velocity.
 - (vi) direction of wave propagation
- [10 marks]
- (b) Displacement equation of a stationary wave of the string is given by
$$y = 5 \cos(0.1\pi x) \sin(50\pi t)$$
where x and y is in m and t is in s. Write the equation for the both progressive waves travelling along the string.
- [2 marks]
- (c) A transverse wave on a taut copper wire has length 1.2 m and mass 0.21 kg. It travels at speed 196 m s^{-1} . Calculate the
- (i) mass per unit length of the cooper wire.
 - (ii) tension on the cooper wire.
- [4 marks]
- 2 (a) A charge of 50 C flows through a silver wire. Calculate the
- (i) number of electrons
 - (ii) amount of current
- flow through a cross-sectional area of the silver wire in 20 s.
- [3 marks]
- (b) A current of 0.5 A flows in a wire of length 2.0 m and diameter 0.15 mm has resistivity of $5.26 \times 10^{-7} \Omega \text{ m}$ at room temperature 25°C . Calculate the
- (i) resistance of the wire.
 - (ii) potential difference across the wire.
- [4 marks]

(c)

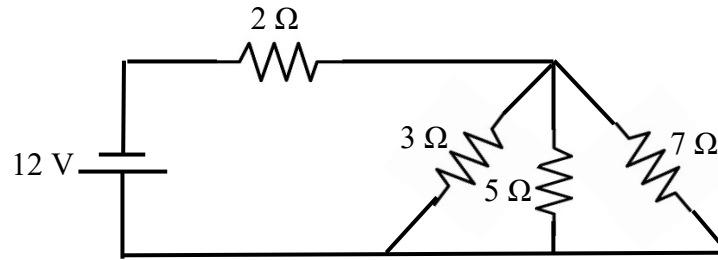
**FIGURE 2**

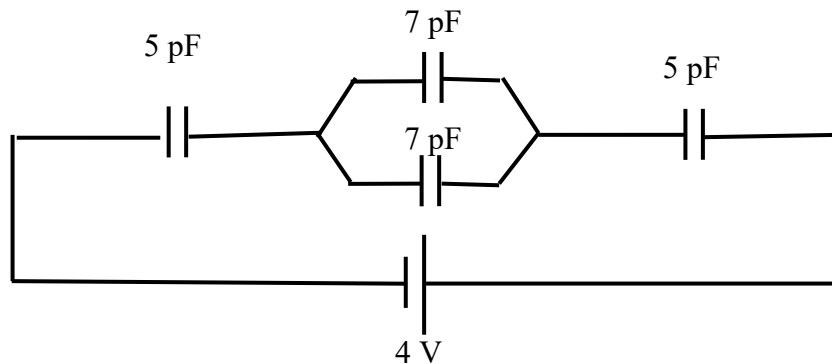
FIGURE 2 shows the 12 V of power supply are connecting with four resistors. Calculate the

- (i) effective resistance of the circuit.
- (ii) potential difference across 5 Ω resistor.

[5 marks]

3

(a)

**FIGURE 3**

Four capacitors are connected to a battery of 4 V as shown in **FIGURE 3**. Calculate the

- (i) equivalent capacitance.
- (ii) total charge of the circuit.

[3 marks]

- (b) A capacitor of capacitance 2000 μF is charged using a battery of 6 V. The charged capacitor is then discharged through a 100 kΩ resistor. Calculate the

- (i) time constant.
- (ii) maximum charge stored in the capacitor.
- (iii) remained in the capacitor after 150 s.

[6 marks]

4

(a)

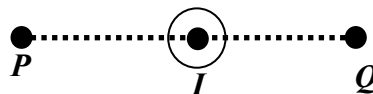
**FIGURE 4(a)**

FIGURE 4(a) shows the top view of a straight current-carrying conductor. Sketch the magnetic field lines produced by the conductor. Label the direction of magnetic field at point **P** and **Q**.

[2 marks]

(b)

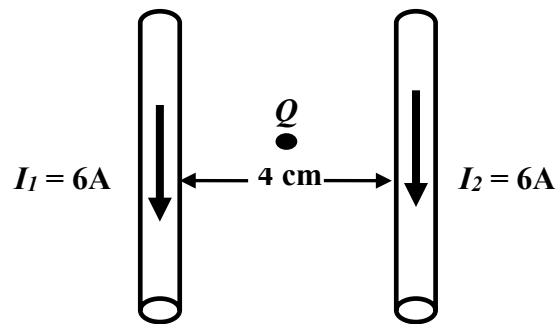


FIGURE 4(b)

FIGURE 4(b) shows two long parallel wires, each carrying a 6 A current in opposite direction. Determine the magnetic field produced by wire 1 at midpoint, **Q**.

[4 marks]

- (c)
- (i) A 0.5 magnetic field is directed parallel to the plane of a circular loop of radius 0.3 m. Calculate the magnetic flux through the loop.
 - (ii) A coil of 150 turns with an area 0.3 cm^2 is placed in a changing magnetic field. The rate of change of magnetic field is 1.25 T s^{-1} . Calculate the induced emf in the coil.
 - (iii) A 100 turns coil with cross-sectional area of $6.0 \times 10^{-3} \text{ m}^2$ is placed in a uniform magnetic field of 1.8 T. Calculate the magnetic flux linkage in the coil when the plane of the coil makes an angle of 50° to the magnetic field.
 - (iv) A coil of wire containing 500 circular loop with radius 3.5 cm is placed in a magnetic field that decreases at the rate of 0.3 T s^{-1} . Calculate the

magnitude of the induced emf if the plane of the coil is perpendicular to the magnetic field.

[8 marks]

- 5 (a) An object of height 40.0 cm is placed 3.0 m from a convex mirror of a focal length 0.4 m.

- (i) Sketch a ray diagram to show the formation of the image.
- (ii) Calculate the image distance.

[5 marks]

- (b) Sofea looks at her face in a makeup mirror. When she is 20.0 cm away from the mirror, her face image is magnified 1.2 times.

- (i) Calculate the image distance.
- (ii) Calculate the radius of curvature of the mirror.

[4 marks]

- (c) An object is placed at 60 cm from a converging lens with a focal length of 15 cm.

- (i) Determine the magnification of the image.
- (ii) State three characteristics of the image.

[5 marks]

- 6 (a) Two parallel slits separated by 0.20 mm are illuminated by 546 nm green light. The interference pattern is observed on a screen 1.5 m away from the slits. Calculate the

- (i) position of the fourth order maximum.
- (ii) position of the third order minimum.
- (iii) distance between the second and the fifth dark fringe.

[9 marks]

- (b) A double-slit experiment is set up using a laser with a wavelength of 632 nm. The distance between the two slits is 0.5 mm, and the distance from the slits to the screen is 2 m.

- (i) What is the distance between adjacent bright fringes on the screen?
- (ii) How does the distance between adjacent bright fringes change if the distance between the slits is decreased? Explain your answer.
- (iii) Suggest one way how to obtain the wider interference pattern.

[5 marks]